MATPMD4 Assignment 1 (Stochastic Processes)

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Module: MATPMD4
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Instructions

I confirm that I have read and understood the instructions on the cover page.

In [1]:
```
# Required Imports
# !pip install numpy matplotlib scipy
import numpy as np
import matplotlib.pyplot as plt
from scipy import stats

# Set precision for cleaner output
np.set_printoptions(precision=4, suppress=True)
```

Question 2

2(a) Calculating x

Reasoning: The matrix P is a **stochastic matrix**. A fundamental property of stochastic matrices is that the sum of probabilities in any given row must equal exactly 1. Mathematically, this is because the process must transition to *some* state (including potentially staying in the same state) in the next time step.

$$\sum_j P_{ij} = 1 \quad \forall i$$

For the third row (State C), we have:

$$0.16 + 0.16 + 0.22 + x + 0.26 = 1$$

$$0.80 + x = 1 \implies x = 0.2$$

In [2]:
```
# Row C is [0.16, 0.16, 0.22, x, 0.26]
# Stochastic matrix property: sum of probabilities in a row must equal 1.
x = 1.0 - (0.16 + 0.16 + 0.22 + 0.26)
x = round(x, 2)  # Round to avoid floating point errors

print(f"\nCalculated x: {x}")

# Define the Transition Matrix P
# Rows: A, B, C, D, E
P = np.array([
    [0.21, 0.07, 0.15, 0.11, 0.46],
```

```
    [0.00, 1.00, 0.00, 0.00, 0.00],
    [0.16, 0.16, 0.22,    x, 0.26],
    [0.00, 0.00, 0.00, 1.00, 0.00],
    [0.21, 0.27, 0.18, 0.24, 0.10]
])

print("\nTransition Matrix P:\n", P)
```

```
Calculated x: 0.2

Transition Matrix P:
 [[0.21 0.07 0.15 0.11 0.46]
 [0.   1.   0.   0.   0.  ]
 [0.16 0.16 0.22 0.2  0.26]
 [0.   0.   0.   1.   0.  ]
 [0.21 0.27 0.18 0.24 0.1 ]]
```

State Transition Diagram

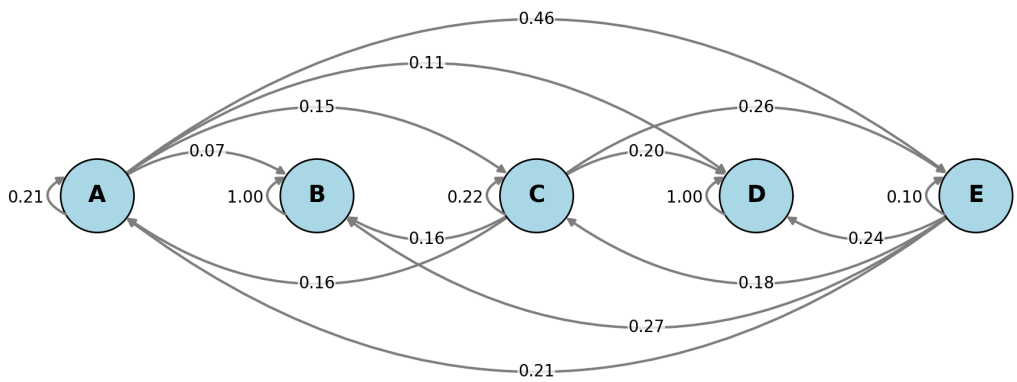


Figure 1: State transition diagram showing transition probabilities P_{ij} for states A-E.

2(b) Distribution after 3 Generations

Observation:
As the number of generations increases, the probability values in the state vector **pi** change significantly:

- The probabilities for states **A** and **C** are consistently **decreasing**.
- The probability for state **E increases initially** (due to inflow from A and C) before eventually **decreasing**.
- The probabilities for states **B** and **D** are consistently **increasing**.

Explanation:
This behavior is expected because **B** and **D** are **absorbing states**.

- Looking at the transition matrix P (and the diagram), states **B** and **D** have a **self-loop probability of 1.0** ($P_{BB} = 1, P_{DD} = 1$).
- This means once the process enters state B or D, it cannot leave.

- States **A**, **C**, and **E** are **transient states**, meaning the process will eventually leave them and get trapped in B or D.
- Therefore, as $n \to \infty$, the total probability of being in **A, C, or E drops to 0**, while the total probability **accumulates** entirely into states **B** and **D**.

In [3]:
```
# Initial distribution: A(13%), B(24%), C(32%), D(28%), E(3%)
pi_0 = np.array([0.13, 0.24, 0.32, 0.28, 0.03])
print(f"Gen 0 Distribution: {pi_0}")

# Generation 1
# Using @ for matrix multiplication
pi_1 = pi_0 @ P
print(f"\nGen 1 Distribution: {pi_1}")

# Generation 2
pi_2 = pi_1 @ P
print(f"\nGen 2 Distribution: {pi_2}")

# Generation 3
pi_3 = pi_2 @ P
print(f"\nGen 3 Distribution: {pi_3}")
```

```
Gen 0 Distribution: [0.13 0.24 0.32 0.28 0.03]

Gen 1 Distribution: [0.0848 0.3084 0.0953 0.3655 0.146 ]

Gen 2 Distribution: [0.0637 0.369  0.06   0.4289 0.0784]

Gen 3 Distribution: [0.0394 0.4042 0.0369 0.4667 0.0527]
```

2(c) Canonical Form

We rewrite P in canonical form by reordering the states so that absorbing states appear first, followed by transient states.

- **Absorbing States:** B, D
- **Transient States:** A, C, E

The canonical form is partitioned as:

$$P = \begin{pmatrix} I & 0 \\ R & Q \end{pmatrix}$$

Where Q represents transitions between transient states, and R represents transitions from transient to absorbing states.

In [4]:
```
# Current State Order: A(0), B(1), C(2), D(3), E(4)
# Absorbing States: B(1), D(3)
# Transient States: A(0), C(2), E(4)

# The new order to group Absorbing first, then Transient.
# New Order: B, D, A, C, E
canonical_order = [1, 3, 0, 2, 4]

# Move Row B to top & Row D to second to top
P_rows_sorted = P[canonical_order, :]
```

```python
# Move Col B to left, Col D to second to left
P_canonical = P_rows_sorted[:, canonical_order]

print("Canonical Matrix P_canonical:\n", P_canonical)

# Structure of Canonical Form:
# | I  0 |  (Absorbing rows)
# | R  Q |  (Transient rows)

# 2 absorbing states and 3 transient states.
num_absorbing = 2
num_transient = 3

# Q: Transient to Transient (Bottom-Right block)
# Take rows from index 2 to end, and columns from index 2 to end
Q = P_canonical[num_absorbing:, num_absorbing:]

print("\nMatrix Q (Transient -> Transient):\n", Q)

# R: Transient to Absorbing (Bottom-Left block)
# Take rows from index 2 to end, and columns from index 0 to 2
R = P_canonical[num_absorbing:, :num_absorbing]

print("\nMatrix R (Transient -> Absorbing):\n", R)
```

```
Canonical Matrix P_canonical:
 [[1.   0.   0.   0.   0.  ]
 [0.   1.   0.   0.   0.  ]
 [0.07 0.11 0.21 0.15 0.46]
 [0.16 0.2  0.16 0.22 0.26]
 [0.27 0.24 0.21 0.18 0.1 ]]

Matrix Q (Transient -> Transient):
 [[0.21 0.15 0.46]
 [0.16 0.22 0.26]
 [0.21 0.18 0.1 ]]

Matrix R (Transient -> Absorbing):
 [[0.07 0.11]
 [0.16 0.2 ]
 [0.27 0.24]]
```

2(d) Fundamental Matrix N

The fundamental matrix N gives the expected number of times the process is in transient state j, given it started in transient state i.

Formula:

$$N = (I - Q)^{-1}$$

Components:

- I: Identity matrix of size equivalent to the number of transient states (3x3).
- Q: The matrix describing probabilities of moving between transient states (*A*, *C*, *E*).

In [5]:
```
I = np.eye(3)
N = np.linalg.inv(I - Q)

print("Fundamental Matrix N:\n", N)
```

```
Fundamental Matrix N:
 [[1.6412 0.5456 0.9964]
 [0.4975 1.539  0.6989]
 [0.4824 0.4351 1.4834]]
```

2(e) Mean Time to Absorption

This calculates the expected number of steps before the process hits *any* absorbing state, given it starts in a specific transient state.

Formula:

$$M = N\mathbf{1}$$

Components:

- N: The fundamental matrix calculated in 2(d).
- $\mathbf{1}$: A column vector of ones.
- Summing the rows of N (which is what multiplying by $\mathbf{1}$ does) adds up the expected visits to *all* transient states, giving the total lifetime of the process.

In [6]:
```
# Formula: M = N @ 1 (column vector of ones)
ones = np.ones((3, 1))
M = N @ ones

print("Mean steps to absorption (from A, C, E):")
print(M)
```

```
Mean steps to absorption (from A, C, E):
[[3.1832]
 [2.7353]
 [2.4009]]
```

2(f) Absorption Probabilities

This calculates the probability that the process will be absorbed into a specific absorbing state (B or D).

Formula:

$$B = NR$$

Components:

- N: The fundamental matrix representing time spent in transient states.
- R: The matrix of probabilities of moving from a transient state directly to an absorbing state.
- Multiplying N by R aggregates the probabilities of all paths leading to a specific absorbing state.

In [7]:
```
# Formula: B = N @ R
B = N @ R

print("Absorption Probabilities (Rows: A, C, E | Cols: B, D):")
print(B)
```

```
Absorption Probabilities (Rows: A, C, E | Cols: B, D):
[[0.4712 0.5288]
 [0.4698 0.5302]
 [0.5039 0.4961]]
```

Question 3

3(a) Parameters and Expected Behaviour

Student Number: 3539054

- $\alpha = 353/100 = 3.53$
- $\beta = 054/100 = 0.54$

Random Walk Equation:

$$dS = a(x,t)dt + b(x,t)dW$$
$$a(x,t) = (\frac{1}{\alpha})\sin(x)\cos(t) = 0.283\sin(x)\cos(t)$$
$$b(x,t) = (\frac{1}{\beta}) = 1.852$$
$$dS = 0.283\sin(x)\cos(t)dt + 1.852dW$$

Expected Trajectory:

The trajectory should be a noisy path that roughly oscillates around a central band over time, with no long-term upward or downward trend.

Justification:

Deduce the behaviour by analysing the terms $a(x,t)$ and $b(x,t)$ separately:

1. **The Oscillatory Drift $a(x,t)$:**
   - The $\cos(t)$ component alternates smoothly between 1 and -1. This means the deterministic push of the walk periodically reverses direction.
   - The $\sin(x)$ term means the strength of that push depends on where the walker currently is. If x is at a multiple of π, the sine is 0, meaning the drift disappears entirely.
   - Because the drift inherently cancels itself out over time, the walk cannot sustain a long-term upward or downward trend. It is forced into a periodic oscillation.
2. **The Dominant Noise $b(x,t)$:**
   - The magnitude of this noise (1.852) is significantly larger than the maximum possible magnitude of the drift (0.283).

- While the underlying drift attempts to create a smooth, bounded oscillation, the overpowering noise will make the actual trajectory appear highly erratic and heavily randomised step-to-step.

3(b) Identifying and Fixing the Bug

The Bug:
The error lies in how dW is simulated. The code implements it as `np.random.normal(0,1)`.

- In the Generalised Wiener Process, dW must be drawn from a normal distribution with a mean of 0 and a **variance of** dt (meaning a standard deviation of $\sqrt{dt}$).
- By using `np.random.normal(0,1)`, the code generates noise with a constant variance of 1, regardless of the time step size.
- This incorrectly causes the diffusion term to scale independently of time, overpowering the simulation.

The Fix:
To fix this, scale the standard normal random variable by $\sqrt{dt}$.

```
x = 0
for i in range(N):
    t = i * dt
    x = x + (1/alpha)*np.sin(x)*np.cos(t)*dt +
(1/beta)*np.random.normal(0, 1)*np.sqrt(dt)
```

3(c) Explanation of Differing Realisations

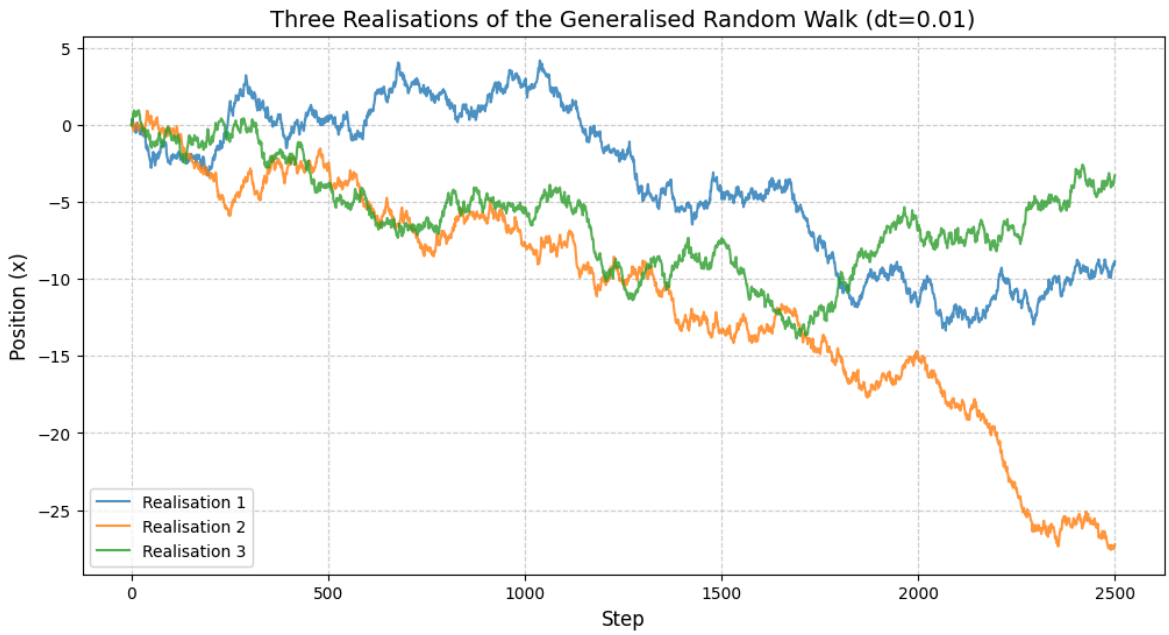
As shown in the plot below, the three realisations diverge from one another.

Why they differ:
Even though all three walks start at $x = 0$ and follow the exact same governing stochastic differential equation, they differ because of the stochastic diffusion term, $b(x,t)dW$.

- At every single time step dt, the code draws a completely independent, random value from a normal distribution to simulate dW.
- Over the course of 2500 steps, these thousands of independent random noise accumulate.
- Because the noise term is the dominant force in this specific equation (as explained in 3a), the compounding effect of these random draws causes the paths to deviate entirely from one another, creating unique trajectories every time the simulation is run.

In [8]:
```
np.random.seed(3)

# Parameters
alpha = 3.53
beta = 0.54
dt = 0.01
```

```
N = 2500

plt.figure(figsize=(12, 6))

# Generate 3 different realisations
for walk in range(3):
    x = 0
    path = [x]

    for i in range(N):
        t = i * dt
        dx = (1/alpha) * np.sin(x) * np.cos(t) * dt + (1/beta) * np.random.norma
        x = x + dx
        path.append(x)

    plt.plot(path, alpha=0.8, linewidth=1.5, label=f'Realisation {walk + 1}')

plt.title('Three Realisations of the Generalised Random Walk (dt=0.01)', fontsiz
plt.xlabel('Step', fontsize=12)
plt.ylabel('Position (x)', fontsize=12)
plt.legend(loc='lower left')
plt.grid(True, linestyle='--', alpha=0.6)
plt.show()
```



3(d) Mean Position Analysis

Results:

The code below displays the mean final position for $dt = 0.05, 0.1, 0.2, 0.4$. Across all tested values of dt, the mean position remains close to 0.

Explanation:

These results are expected due to the mathematical structure of the stochastic differential equation $dS = a(x,t)dt + b(x,t)dW$.

1. **Symmetric Noise:** The random diffusion term is driven by a standard normal distribution centred around 0. Over 1000 independent walks, the positive and negative random noise average out to zero.

2. **Symmetric Drift:** The deterministic drift term includes $\sin(x)$. Since the walk starts at $x = 0$, the initial drift is exactly 0. Furthermore, the sine function is an odd function ($\sin(-x) = -\sin(x)$). This means if the noise pushes the walk into positive territory, the drift behaves one way, but if it pushes it identically into negative territory, the drift behaves in the exact opposite way.

Conclusion:
Because both the noise and the spatial drift behave symmetrically around the starting point of $x = 0$, there is no mathematical bias pushing the aggregate population of walks predominantly upward or downward. Therefore, the expected mean position of many independent walks will always be approximately 0, regardless of the time step size dt.

In [9]:
```
np.random.seed(0)

num_walks = 1000
dt_values = [0.05, 0.1, 0.2, 0.4]

print(f"Mean final position after {N} steps (averaged over {num_walks} walks):")
print("-" * 64)

for dt in dt_values:
    # Initialize a vector 'x' to track the positions of all 1000 walks.
    # x_0 = 0 for all paths.
    x = np.zeros(num_walks)

    for i in range(N):
        t = i * dt

        # Drift: a(x,t)*dt
        # Passing the vector 'x' into np.sin() calculates the spatial drift
        # for all 1000 walks simultaneously (element-wise).
        drift = (1/alpha) * np.sin(x) * np.cos(t) * dt

        # Noise: b(x,t)*dW
        # size=num_walks forces NumPy to draw 1000 independent samples.
        # Every single walk receives its own independent random noise.
        noise = (1/beta) * np.random.normal(0, 1, size=num_walks) * np.sqrt(dt)

        # This vector addition updates the positions of all 1000 walks at once,
        # mathematically identical to running 1000 separate loops, but faster.
        x = x + drift + noise

    mean_position = np.mean(x)
    print(f"dt = {dt} | Mean Final Position = {mean_position:.4f}")
```

```
Mean final position after 2500 steps (averaged over 1000 walks):
----------------------------------------------------------------
dt = 0.05 | Mean Final Position = 0.3550
dt = 0.1 | Mean Final Position = -0.0650
dt = 0.2 | Mean Final Position = 1.6930
dt = 0.4 | Mean Final Position = 0.7062
```

3(e) Histogram and Normality Test

Visual Assessment:
The histogram of the final positions for 5000 walks visually resembles a normal

distribution (a bell-shaped curve) centred around 0. This central clustering aligns with the findings in 3(d).

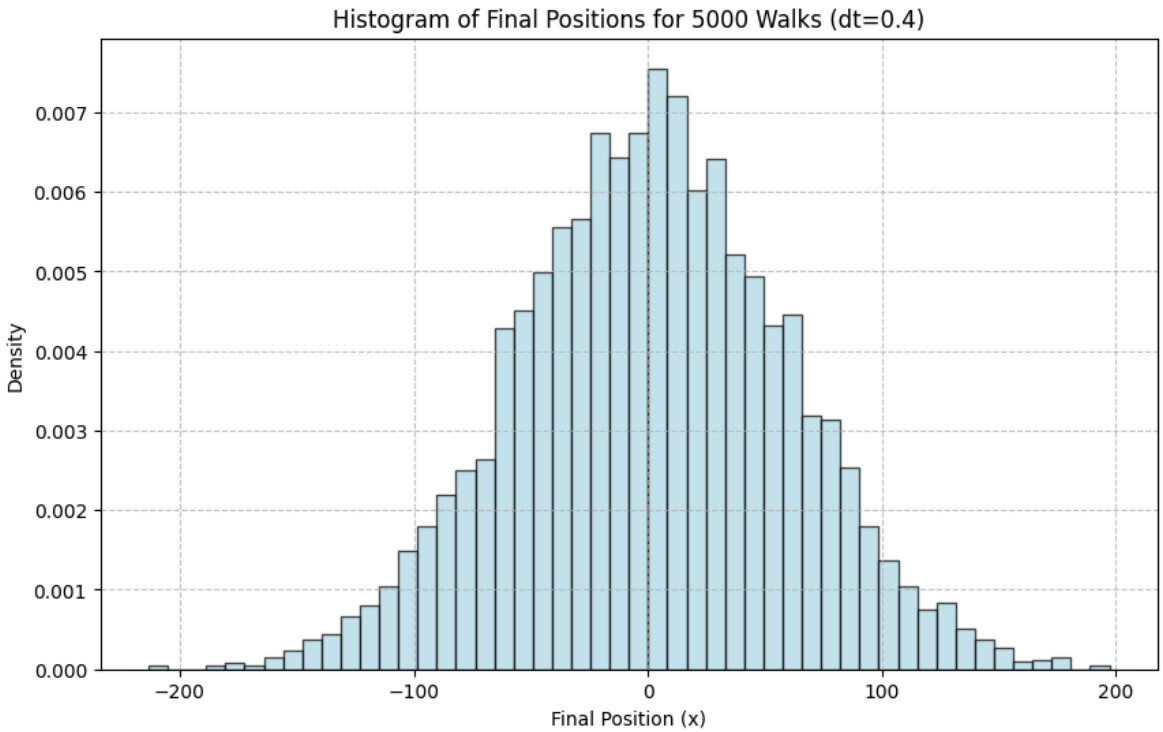
Statistical Testing:
Used D'Agostino's K-squared test (`scipy.stats.normaltest`) to formally test if the data is normally distributed.

- **Null Hypothesis:** It is a normal distribution
- **Test Statistic:** 0.0979
- **p-value:** 0.9522

Conclusion:
Using a standard significance level of $\alpha_{sig} = 0.05$, the p-value is larger than the threshold ($p > 0.05$). Therefore, we fail to reject the null hypothesis. This supports the visual assessment that it is a normal distribution.

In [10]:
```
np.random.seed(3)

num_walks = 5000

# Initialize a vector 'x' to track the positions of all walks.
x = np.zeros(num_walks)

for i in range(N):
    t = i * dt
    drift = (1/alpha) * np.sin(x) * np.cos(t) * dt
    noise = (1/beta) * np.random.normal(0, 1, size=num_walks) * np.sqrt(dt)
    x = x + drift + noise
```

In [11]:
```
# Plotting the histogram
plt.figure(figsize=(10, 6))
# Using density=True to normalize the histogram,
# making it easier to compare to a standard curve
plt.hist(
    x, bins=50, density=True, alpha=0.7,
    color='lightblue', edgecolor='black'
)
plt.title(f'Histogram of Final Positions for {num_walks} Walks (dt={dt})')
plt.xlabel('Final Position (x)')
plt.ylabel('Density')
plt.grid(True, linestyle='--', alpha=0.7)
plt.show()
```



In [15]:
```
# Run D'Agostino's K-squared test for normality
stat, p_value = stats.normaltest(x)
print(f"Normality Test Statistic: {stat:.4f}")
print(f"p-value: {p_value:.4f}")
```

```
Normality Test Statistic: 0.0979
p-value: 0.9522
```

3(f) Probability of $|x| \leq 2$

Empirical Calculation:
Using 5000 realisations generated in 3(e), the estimated probability is:

$$P(|x| \leq 2) \approx 0.0282$$

Dependency Analysis:
Yes, this probability depends on both the time step size (dt) and the number of steps (N).

1. **The Mathematical Relationship:** In a stochastic process like this one, the variance of the distribution grows linearly with the total elapsed time, T. The total time is defined as $T = N \times dt$.

2. **Impact of dt and N:** Because the variance is proportional to $(N \times dt)$, increasing either the number of steps or the size of the time step will increase the total variance of the final positions.

Conclusion:
As variance increases, the bell curve of final positions widens and flattens out. A wider distribution means a smaller proportion of the total probability mass remains clustered near the origin. Therefore, increasing either dt or N will decrease the probability that a walk ends up within the bounds of $[-2, 2]$.

In [13]:
```
# Using the array 'x' (final positions of 5000 walks) generated in 3(e)

# Count how many walks ended up between -2 and 2
walks_in_range = np.sum((x >= -2) & (x <= 2))

# Calculate the empirical probability
probability = walks_in_range / len(x)

print(f"Number of walks ending in range [-2, 2]: {walks_in_range} out of {len(x)
print(f"Empirical Probability P(|x| <= 2): {probability:.4f}")
```

```
Number of walks ending in range [-2, 2]: 141 out of 5000
Empirical Probability P(|x| <= 2): 0.0282
```